An Automated Alert System for Monitoring the Hygiene in Restaurants using Machine Vision

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***Abstract*—Hygiene violations in restaurant kitchens pose significant risks to public health, necessitating effective monitoring and compliance systems. This paper introduces a sophisticated system integrating machine learning, computer vision, and cloud technologies to monitor and enforce hygiene standards in restaurant kitchens. The system uses a YOLOv8 machine learning model trained on a specialized dataset to detect hygiene violations such as improper attire (apron, gloves, hairnet), presence of pests (lizard, rat, cockroach), and adherence to hygiene protocols. Real-time monitoring is facilitated through connected cameras strategically placed in kitchen areas, allowing instant detection and notification of violations. The system's capabilities are improved by the integration with Firebase services, which include Cloud Storage and Database. The Firestore database securely stores violation details and sends out real-time alerts to ensure prompt resolution of such problems. The system achieves an impressive accuracy rate of 89% in identifying and categorizing hygiene violations, ensuring adherence to food safety regulations and safeguarding public health in restaurant environments.**

Keywords—Hygiene Violation, Machine Vision, Firebase, YOLOv8.

I. Introduction

Restaurants are crucial to offering high-quality food and guaranteeing patrons a secure dining environment. However maintaining ongoing vigilance over cleanliness measures can be difficult, especially in larger applications. Cleaning may be challenging to maintain when manual inspections and monitoring procedures are labour-intensive and prone to human mistakes. The use of machine vision [1] technologies to monitor hygiene standards has become more significant in response to these problems. With the use of this technology, visual data from cameras placed in restaurant kitchens may be monitored in real-time. The system may automatically monitor a variety of hygiene issues, such as gloves, an apron, and the cleanliness of the work area, by merging computer vision algorithms and artificial intelligence approaches.

To ensure the health and well-being of the people, restaurants today must meet strict standards of safety and cleanliness. Staff member's hygiene habits, frequent cleaning and sanitization, and comprehensive food handling training are all essential components of efficient hygiene monitoring in restaurants [2].

The goal of machine vision is to provide image-based automatic inspection and analysis for a range of industrial functions, such as automatic inspection, process control, and robot guidance [3]. Applications, software hardware modules, integrated systems, processes, and knowledge are all extensively covered. Object identification and categorization using Convolutional Neural Networks (CNNs) [4] such as YOLOv8, R-CNN, Fast R-CNN, etc. It is feasible to train the dataset and identify the items by utilizing these techniques [5].

The current approaches for detecting hygiene have a great deal of drawbacks. These consist of labour-intensive and manual procedures, a deficiency in real-time monitoring, timely alerts and training, inadequate record-keeping, a higher chance of equipment malfunction, non-compliance hazards, and a lack of scalability. One of the key drawbacks is the reliance on the current methods for manual checks and paper-based reports, which can be laborious and prone to human mistakes. The risk of foodborne illness may increase, and inspection completion time may be prolonged. Another issue is inadequate training, as managers and employees with less experience may not receive the necessary instruction in food hygiene procedures. The disadvantages of existing systems include that, the image captured might be sometimes affected by lighting and environmental factors of the kitchen and detection accuracy for smaller objects is less.

So, the problem statement is to develop an automated hygiene detection and alert system for restaurant kitchens, ensuring adherence to cleanliness standards and immediate response to hygiene violations.

The objectives of the proposed system include developing an automated hygiene monitoring system, to alert the respective authorities and developing an Android app for alerts.

This technology offers a practical way to assist restaurant kitchens in putting in place an automated hygiene detection and alert system. The major goals of the system are to keep an eye on and uphold adherence to cleanliness standards, identify hygiene infractions quickly, and guarantee a timely response

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when violations occur. The system takes advantage of technologies that are necessary to aid with automation, like machine vision. Given that it operates on smartphones, this notion has a very broad application. The technology offers effective, in-the-moment activity monitoring and analysis for the kitchen.

II. Literature Review

First, To understand more about the proposed system, the summary of the most relevant 15 papers is set down below.

Tao et al. [6] proposed a model for the implementation of a deep learning-based kitchen food safety standards detection system equipped with monitoring to stop safety and hygiene issues brought on by irregular activities. They have used a method to automatically detect kitchen staff violations using deep learning. The technology uses a camera in the kitchen to scan the recorded video stream for several unusual behaviours that could compromise the safety of the food being prepared. The model was implemented in the field of food safety, within an Edge Computing Service Infrastructure. The advantage of this system is in mastering pattern recognition, and enhancing food safety safeguards. However, despite the accuracy of image recognition, the model has some inherent limitations. It gets difficult when many of the photos it needs are unavailable because advanced techniques strongly rely on them.

Zhonghui and Juncheng [7] proposed a model to improve kitchen safety, they researched applying computer image recognition technology to kitchen hygiene inspections. They have developed a kitchen hygiene inspection security system that uses computer image recognition technology to monitor the hygiene management of users' kitchens. Computer image recognition technology was used for the hygiene inspection of the kitchen. To assess the degree of contamination, measurements of kitchen contaminants are compared with reference values. Support Vector Machine algorithm is used for image recognition and classification. Texture features are selected as the feature vector for training the SVM. The merits of the model include recognition of indoor scenes and improved health and safety of the user. The demerits of the implemented model include a reduction in accuracy due to factors like noise, blurring, or reduced image quality.

Nayar et al. [8] have proposed a model for pest detection. They have chosen pest traits like size, shape, colour, and texture by hand using traditional machine vision approaches. By tackling the difficulties associated with pest detection, the model seeks to increase the precision and effectiveness of pest detection techniques. The model was implemented in the domain of pest detection. For large-scale, multi-class, dense, and microscopic pest identification and counting, it integrates the YOLOv7 model. The model proposed has several merits. Firstly, it incorporates the Adan optimizer, which enhances the detection accuracy of corn pests. This suggests that the model is effective in accurately identifying and detecting corn pests. However, the model fails to cover a broader range of pests, the resources are limited, and less accuracy and diversification due to the usage of the smaller dataset.

Galkin et al. [9] proposed a model for a vision-based hygiene monitoring system that uses deep learning to evaluate handwashing processes in real time. The model uses the ResNet-18 neural network, which can identify six different hand poses, which helps to effectively monitor and evaluate the handwashing process. They utilized deep learning techniques to develop their model, which allows for real-time assessment of hand hygiene practices. This approach allows their model to analyze video footage and assess the key steps and techniques involved in handwashing. They have implemented their model in the domain of hygiene monitoring. The approach they use has several benefits, providing immediate feedback on handwashing methods helping in making improvements in hygiene practices and preventing the spread of diseases and infections. It also offers a non-intrusive and automated solution for monitoring hand hygiene. The vision-based system eliminates the need for manual observation or self-reporting. The method has the weakness of not mentioning problems caused by data security and privacy issues.

Sathyamurthy et al. [10] proposed a model for real-time face mask detection using Tiny YOLO V4 to prevent the spread of the pandemic. They have utilized the concepts and techniques of object detection and the Tiny-YOLO V4 architecture in their model. The proposed model can efficiently detect the presence or absence of a face mask despite the environmental conditions or the location of the camera. The YOLO V4 algorithm discovers many object classes at lightning speed without using any hardware resources. They have implemented their model in the domain of face mask detection, focusing specifically on real-time scenarios. The merits of the proposed model include its ability to detect face masks in real-time, enabling efficient monitoring and enforcement of mask-wearing policies. The Tiny-YOLO V4 architecture has higher speed and accuracy than other algorithms. The demerit of the model is that errors can be caused due to partial occlusion, variations in lighting conditions, or challenges in differentiating between face masks and other facial coverings.

Raote et al. [11] have proposed a model for ensuring health safety and hygiene on a college campus. They have used object detection algorithms, including MobileNet and SSD architectures, for crowd counting, social distancing, and mask detection. The model was successful in counting the number of people entering and leaving the premises, detecting violations of social distancing rules, and identifying those who were not wearing masks. When deployed on university campuses, this model has proven its effectiveness in real-time monitoring and alerting authorities. The model alerts authorities in real-time if it detects any violations, which can help in immediate action and enhance the safety standards. However, it has certain limitations. The model's reliance on specific threshold values for detecting violations may not perfectly align with real-world variations and also the scope of the model is limited to the campus environment and might require modifications for other settings or scenarios.

Bernabe et al. [12] proposed a model for kitchen safety monitoring. They used the Haar Cascade Classifier technique to resolve the kitchen safety monitoring problem. They address the issue of monitoring and detecting potential safety hazards in a kitchen environment Haar Cascade Classifier is a popular computer vision technique that leverages predefined features to detect objects accurately. It is particularly suitable for real-time applications like kitchen safety monitoring. They have implemented this model for Kitchen Safety Monitoring, where the objective is to detect various safety issues such as gas leaks, fires, and dangerous behaviours. Their model shows promising results in terms of accurate detection and real-time performance. By leveraging the Haar Cascade Classifier, they were able to detect and alert potential safety hazards promptly.

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However, the effectiveness of object detection is significantly influenced by the quality of the training data and the cascade features that are defined. Also, the model's performance may be affected by variations in lighting conditions or occlusions within the kitchen environment.

Shrimali and Teuscher [13] proposed a model for enforcing thorough hand hygiene compliance in medical facilities. To address the issue of hand hygiene compliance in hospital settings, researchers have developed a unique deep learning technique in conjunction with camera and sensor-based devices. By leveraging deep learning techniques, they aimed to improve the accuracy and effectiveness of monitoring hand hygiene practices. This specific method is used in hospital settings where maintaining good hand cleanliness is essential to preventing the spread of disease. Their model demonstrates several merits, including enhanced accuracy in detecting hand hygiene compliance, real-time monitoring capabilities, and the potential to provide valuable feedback to healthcare professionals. The integration of deep learning, cameras, and sensors allows for reliable and continuous monitoring, reducing the reliance on manual observation and improving overall compliance rates. However, their model has certain limitations such as the system's effectiveness might be influenced by factors such as lighting conditions, occlusions, or variations in hand positioning during hand hygiene activities.

Ren et al. [14] have proposed a model for developing a people-counting solution that is more precise and effective. They have used a novel approach based on the You Only Look Once (YOLO) algorithm for resolving the problem of carrying out real-time people counting in congested places. They have implemented this model in planning for cities, crowd control, retail analytics, and security. Their model shows a YOLO-based method that enhances people counting's speed and accuracy, making it highly efficient for applications that work in real-time. The model also demonstrates the effectiveness of YOLO as an object detection system, emphasizing its significant benefits in handling tough circumstances that involve people counting where traditional methods might fail. However, their model fails to address the increased chances of incorrect results in situations with small or partially obstructed objects, in this case, people in a crowd, which is one of the related problems with YOLO-based systems, and as such with this model. This method improves in counting in real- time, but because it does not allow for the analysis of historical data, it is limited in how it can be used for trend-based analysis or long-term planning.

Anly et al. [15] have proposed a model for determining the quality of food with the aid of modem machine learning methods. They have used various methods that rely on the data, including Decision Tree (DT), K-Nearest Neighbour (KNN), Support Vector Machine (SVM), and Artificial Neural Network (ANN) for resolving the huge task of uniform, objective, and automated food quality assessments and predictions. Their model shows a thorough evaluation of various machine learning methods for predicting food quality, providing important field-relevant conclusions. This method promotes automation and objectivity in a largely subjective field, helping in the standardization of food quality evaluation. However, their approach ignores the fact that, in real-world circumstances, logistical and practical challenges may make it difficult to always have access to high-quality sensory input. A thorough analysis of the potential bias in the predictive algorithms that could influence the results of food quality predictions is also lacking in the study.

III. Proposed System

The proposed system consists of a camera, a central coordinating module and a mobile application for detecting hygiene violations in restaurant kitchens. The camera captures real-time images from the kitchen. After analyzing the images, the system identifies the violations and sends an in-app alert message. The system is trained using the YOLOv8 model. Fig.1 shows the development block diagram.

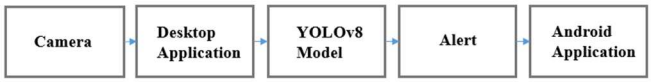

Fig.1. Block Diagram of Proposed Model

The system starts by capturing the image from the kitchen which is then processed for object detection. If a violation is detected, an alert message is sent. If no violation is found, the process loops back to capturing the next image. This cycle repeats until the desired kitchen monitoring is achieved, ensuring a continuous check for potential hygiene concerns.

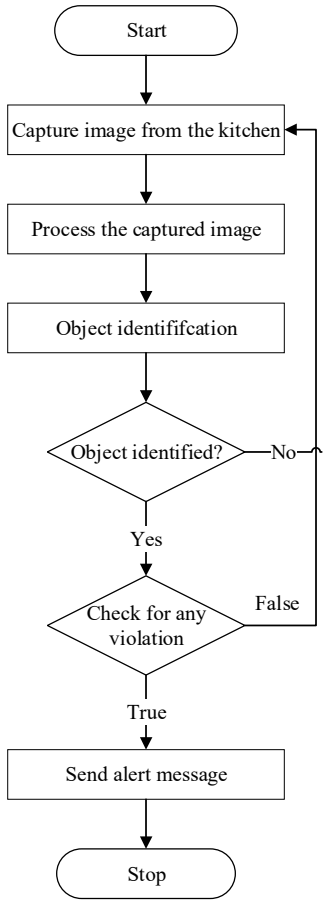

Fig.2. Flow Chart

Fig.2 shows the flow chart of the Hygiene Monitoring System. The system starts by capturing the image from the kitchen which is then processed for object detection. If a violation is detected, an alert message is sent. If no violation is found, the process loops back to capturing the next image. This cycle repeats until the desired kitchen monitoring is achieved, ensuring a continuous check for potential hygiene concerns. The various processes involved in the system are discussed below.

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A. Android Application

The alerts for hygiene violations are provided by the Android app. If a violation is identified, an in-app alert message is sent along with the violation image, date and time. Also, the history is stored in the app. The history can be viewed at any time by the user.

B. Object Detection

The YOLOv8 model is used for object detection. YOLOv8 is the latest version of the YOLO object detection model with improved accuracy and efficiency over the previous versions. The dataset consists of 9 classes: apron, no_apron, hairnet, no_hairnet, gloves, no_gloves, mouse, lizard and cockroach. The model identifies to which class the image captured by the camera belongs. The model shows an accuracy of 89% with higher efficiency.

C. Image Dataset

Dataset is a key aspect of the Machine Learning process and choosing appropriate input data is a crucial part. A wide variety of datasets has been collected from different sources the main being Roboflow Universe along with images that have been acquired from similar sample environments for the hygiene monitoring system to identify any type of hygiene violation occurring in restaurant kitchens. The dataset also contains images captured from the working environment. The image has a resolution of 640×640 pixels in size. A few numbers of images in the dataset were used for testing the model.

D. Data Pre-processing

The dataset consists of 9400 images that need to be pre-processed before training them. A pre-processing approach written in Python resizes all training images in the dataset automatically instead of manually resizing each image, which is time-consuming. All images in the dataset were resized to 640×640 pixels for data consistency. It is done to reduce execution time and simplify the training process. The dataset annotation is a crucial part of data preprocessing in the case of object detection. Annotation is the process of adding tags and labels to raw data that helps to identify various features within the dataset that are essential for the object detection model while training.

E. YOLOv8 Object Detection Algorithm

A popular object detection algorithm in computer vision is called YOLOv8. It is the most advanced algorithm available today. This model is the most recent in the YOLO series. Its excellent precision and capacity for real-time object detection make it more pertinent. By calculating bounding boxes and class probabilities for each grid cell and organizing the information into a grid, YOLOv8 aims to detect objects in images or video frames. The result can be successfully forecasted using this technique in a single forward run through the neural network. Compared to other models, YOLOv8's performance and mAP values are significantly higher. YOLOv8's primary goal is to outperform its predecessors in terms of accuracy and efficiency. With that function, the input image size will be reduced to size 224x224

By predicting bounding boxes and classes of objects directly from an input image, a single neural network can increase efficiency and accuracy while doing away with the need for several detection steps. The YOLOv8 model exhibits good accuracy and quickness as well. Real-time applications can benefit from its enhanced detection speed and high precision, both of which are markedly improved over earlier iterations. To detect objects, YOLOv8 uses a neural network design. The object detection neural network is designed and trained to identify objects, identify the class of items in a picture, and return the names and locations of those objects. Fig.3 shows the workflow diagram of the proposed model.

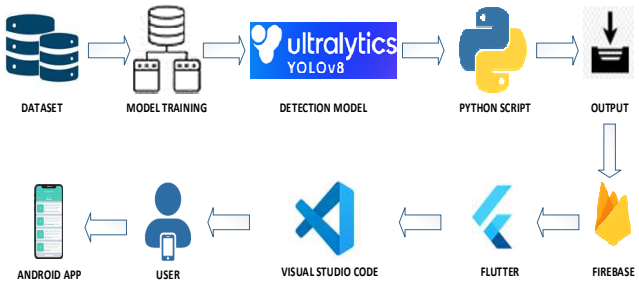

Fig.3. Workflow Diagram

IV. System Implementation

PyTorch plays a crucial part in the machine learning and deep learning domains within the framework of the automated hygiene monitoring system. As a computational engine, PyTorch makes it possible to apply complex object detection models, such as YOLO, which is essential for identifying hygienic violations in restaurant settings. YOLO is used to train the dataset. A dataset was created and it was trained using YOLOv8. The real-time object detection features of YOLOv8 enable the system to quickly and precisely detect hygiene violations in a restaurant kitchen. Its unique approach processes the entire scene in a single pass, providing comprehensive detection and precise localization of multiple hygiene-related objects.

Using YOLO, it is easier and faster to detect the violations and thus send the violations to respective authorities.

a) *Importing Libraries:* Importing necessary libraries for the working of the YOLOv8 including the libraries that are needed for PyTorch and image processing.
b) *Data Collection:* Labeled dataset containing images with annotated bounding boxes for kitchen hygiene violations and pests are collected and the dataset is organized as training and validation sets.
c) *Data Pre-processing:* Images need to be resized to 640x640 pixels, which is given as input to YOLOv8. The annotations are converted into YOLO format.
d) *Build the model:* The YOLOv8 model is built by assembling components for object detection, including a backbone network (CSPDarknet53) for feature extraction and a detection head for predicting bounding box coordinates and class probabilities.
e) *Training Model:* The YOLOv8 model is trained on the annotated dataset using the configured hyperparameters.
f) *Validation:* The validation dataset is used to evaluate the model's performance.
g) *YOLOv8 Model File:* The trained YOLOv8 model weights are used for performing object detection. Here the YOLOv8 model is converted to the PyTorch model for the deployment.

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YOLOv8 uses a combination of supervised and unsupervised learning. First, the input image is divided into cells and boundary boxes are predicted. The most likely bounding boxes for each object in the image are then chosen by YOLOv8 using a non-maxima suppression (NMS) technique to filter out overlapping bounding boxes.

There are nine classes in the dataset which include rat, hairnet, no_hairnet, apron, no_apron, lizard, cockroach, gloves, and no_gloves. The PyTorch architecture-based model is trained on each folder in the dataset, labelling them as 0, 1..., and so on. The output model is then translated to the.pt format. Based on the input image captured by the camera module, each violation is identified. Subsequently, the nine classes in each dataset that have violations are trained, and the prediction rate is assessed.

The Firebase platform is a database for storing and retrieving violations. Hotel managers and food inspectors get an easy workflow as it instantly synchronizes violation facts across all linked devices via the Firestore database. Furthermore, the system connects with the Database's messaging services, which makes sure that notifications and updates about violations reach the Android app quickly, which improves the system's responsiveness.

The Android application functions as the hub for effectively handling violations related to kitchen hygiene. The application serves two different user roles, hotel manager and food inspector. After obtaining violation details from Firebase, food inspectors may easily navigate and examine violations across the hotels, while hotel managers can access individual problems within their hotel to enable targeted monitoring. Complete information about every violation, such as type, date, and time, is provided by the Android application called RestoCare. Due to this improved capability, food inspectors and hotel managers can easily keep informed and take appropriate action based on the details of each violation. Fig.4 shows the Desktop Application view and Fig.5 shows the Android app view. The Desktop Application is used to control the start and stop actions of the camera.

The hygiene violation module is converted into a Desktop Application. This module facilitates the configuration of the hygiene monitoring system by allowing users to select the hotel name and camera input source. It also includes controls to start and stop the process. It serves as the user interface for an advanced application designed to enhance hygiene monitoring in a hospitality environment. The desktop application uses Tkinter, a Python interface to the Tk GUI toolkit, for its robust and user-friendly graphical interface. It plays the role of the central coordinator as in Fig.1. Python serves as the primary programming language due to its extensive libraries and ease of development.

Testing is the procedure used to validate and confirm a developed product. Validation verifies that the product is developed correctly and in the right way, whereas verification checks whether the developed product has all the necessary functionalities. Below are the findings of the external evaluation of the machine learning model and the Android app. The model has been tested manually and the result has been noted.

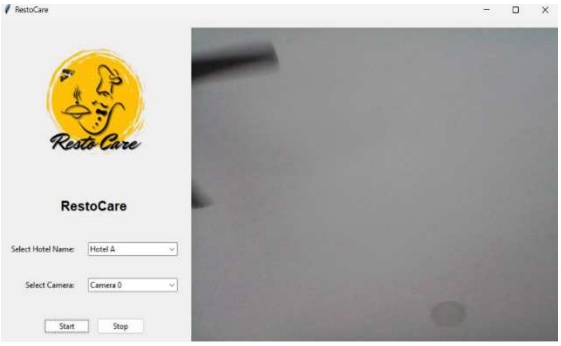

Fig.4. Desktop Application

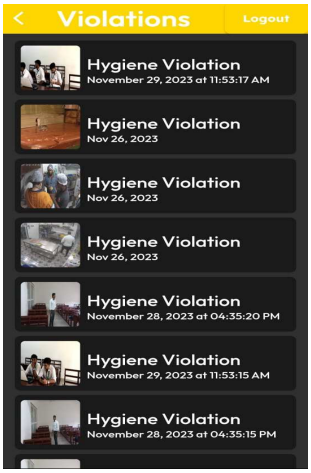

Fig.5. Displaying Violations Details

V. Results

Based on the YOLOv8 object detection algorithm, a kitchen hygiene monitoring system is created that identifies any type of hygiene violation happening in the kitchen. The performance of the model is analyzed using 1123 images of the test set from among the 9400 images. The system showed an overall accuracy of 89%, and there were only a few errors caused due to the variation in image quality of images captured from existing CCTV cameras.

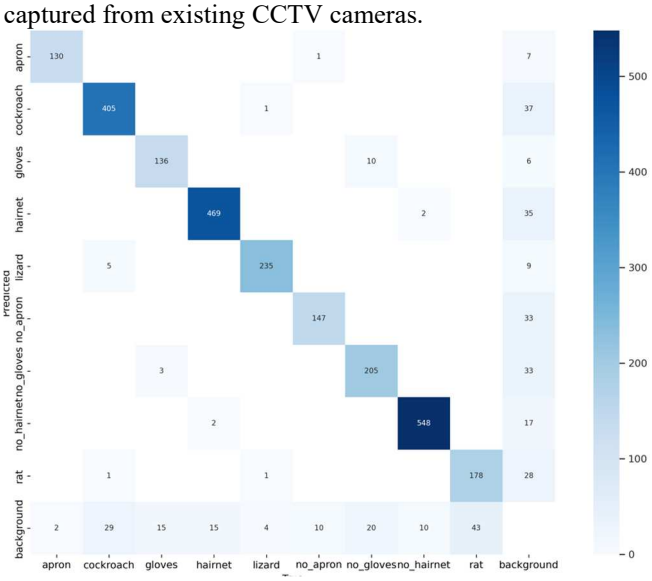

Fig.6. Confusion Matrix

The confusion matrix provides a detailed overview of the model's performance in classifying different hygiene violations as shown in Fig.6. It illustrates the true positives,

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false positives, true negatives, and false negatives for each class, offering insights into the model's strengths and areas for improvement.

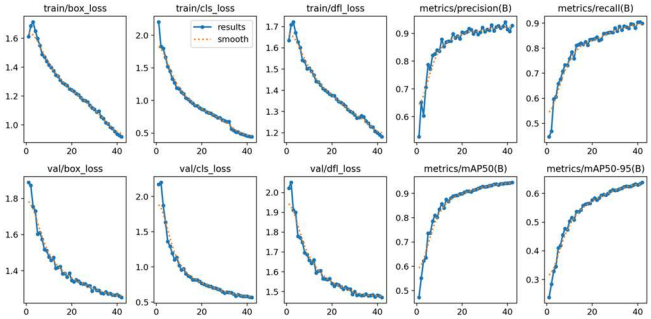

Fig.7. Results Graph

Fig.7. shows the results graph which visually represents the performance metrics of the model across different classes of hygiene violations.

VI. Conclusions

The kitchen hygiene system stands as an important advancement in ensuring food safety and sanitation in commercial kitchens. This model, trained to identify various hygiene violations, exhibits an impressive accuracy of 89%, encompassing crucial elements such as the presence of pests, adherence to safety clothing, and more. The system seamlessly integrates with Firebase services and uses the Firebase Realtime Database for storing violation images and details. Firebase Authentication ensures secure user access, while Firestore plays a crucial role in delivering real-time alerts to the mobile app. The mobile app, designed for user interaction, offers a streamlined experience. The app provides a detailed history of violations, allowing users to review past incidents at their convenience. Focusing on the entire system, from violation detection through Python script to Firebase integration and mobile app functionality, serves as a robust solution for intelligent kitchen hygiene monitoring. This holistic approach, combining advanced technologies and seamless data flow, addresses critical challenges in maintaining high standards of food safety and hygiene in commercial kitchens. The model identifies and rectifies violations promptly but also establishes a foundation for continual improvement and awareness in the realm of kitchen hygiene.

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